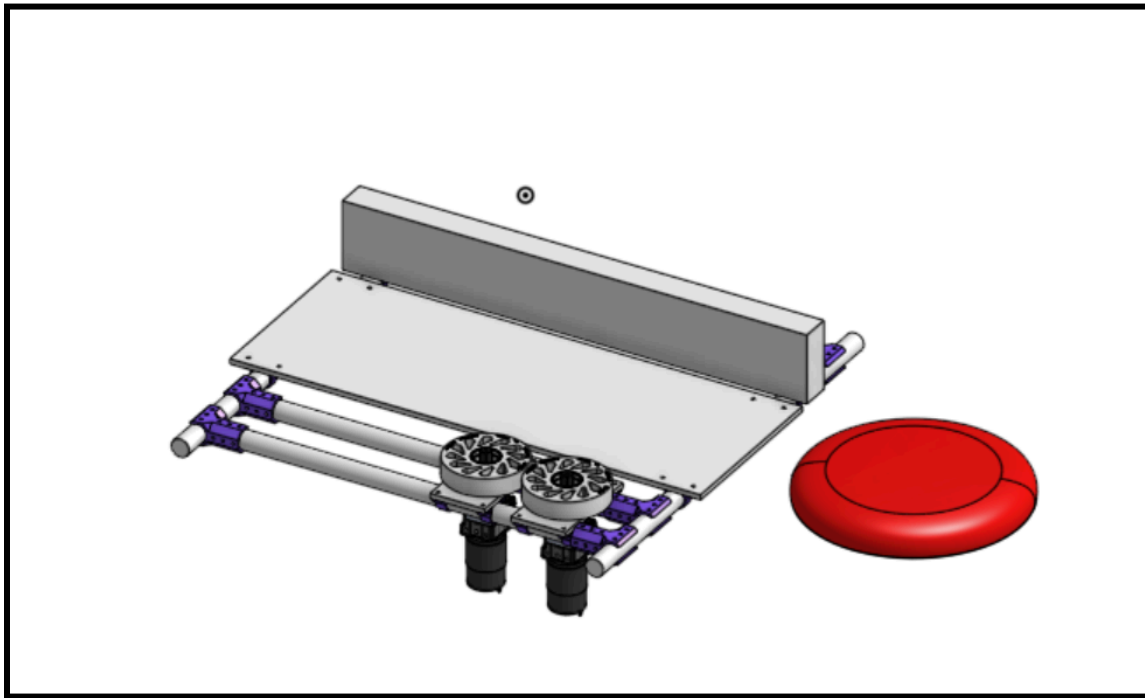


Element G: Construction of a Testable Prototype

The final iteration of the frisbee launcher was built deviating from the original design in order to make it more efficient and cost-effective. For example, instead of utilizing 2 motors to operate the flywheels, we used one motor and used a belt to operate the flywheels at the same velocity. This allows for the operators to easily replace the motor if necessary. In addition the side wall that ensures the frisbee is kept within proper distance to the flywheel ensures that it is modifiable meaning that the wall can be adjusted simply with the use of a screwdriver and a nut driver. This feature ensures that the operator can adjust the wall when needed so that the performance of the launcher is not limited.

CAD:



This image is our CAD model of our frisbee launcher

Bill of Materials(BOM):

Name	Material	Quantity	Cost
Wood Base	Wood Plank	1	*Calculated by slicer
Flywheel	Rubber	2	*Calculated by slicer
Motor Pulley	Aluminum	1	\$15.99
Axle Pulleys	PLA	2	*Calculated by slicer
775 Pro Motor	*N/A	2	\$7.99
Motor Mount	Aluminum	N/A	N/A
Shaft Collars	PLA	4	N/A
Bearings	Steel	4	\$1.99
Axles	Metal	2	\$4.38

Process of Construction of Prototype:

In order to have a solid base for the frisbee shooter, a plank of wood was used with a rubber side barrier used to guide the frisbee into the flywheels. To mount the side support onto the wood plank, the CNC machine was used to make holes into the rubber. From there, screws were used to attach the side piece to the plank and were secured by nuts. The next steps were to mount the motor in which an aluminum piece would go from one side of the base to the other side on the bottom. Two smaller aluminum pieces were sanded and used to raise the longer piece where the motor would be attached. Once the smaller pieces were screwed in, an L-shaped bracket attached the longer piece to the end pieces for the longer piece to be raised at a certain height. Then the motor was attached to the longer aluminum piece using the bracket that originally came with the motor. Next, the holes for the bearings and axles were drilled using a drill press with a 1/4th inch drill bit. In order for the bearings to be secured, electrical tape was wrapped around the rim of the bearing and was pressed into the hole. Now that the bearings were secured into place, the axles were pushed through the bearings and ensured that they weren't loose by screwing in shaft collars at the top and bottom. The 3-D printed pulleys were attached to the ends of each axle and the aluminum pulley was attached to the motor. Then, the belt was wrapped around all the pulleys, connecting the whole flywheel and motor system together.

In addition to the physical hardware, there were mechanical and electrical subsystems used to make the frisbee operational. One motor was used to connect two flywheels using a belt and pulley system. The flywheels were connected to a rod which was secured to the plywood via bearings. These flywheels were also connected to each other through the use of a gear belt. As a result, the two flywheels could now operate as one single unit powered by one single motor. This allowed for a smoother and faster spinning of the flywheels. The motor was attached to the bottom of the frisbee launcher but was still able to power the flywheels which were above the plywood. Thanks to the pulley and belt mechanism, all subsystems were easily connected to become one single system.